

Airframe Technology Master

Checkstress

Lecture 13. Metallic Sizing (Fitting example)

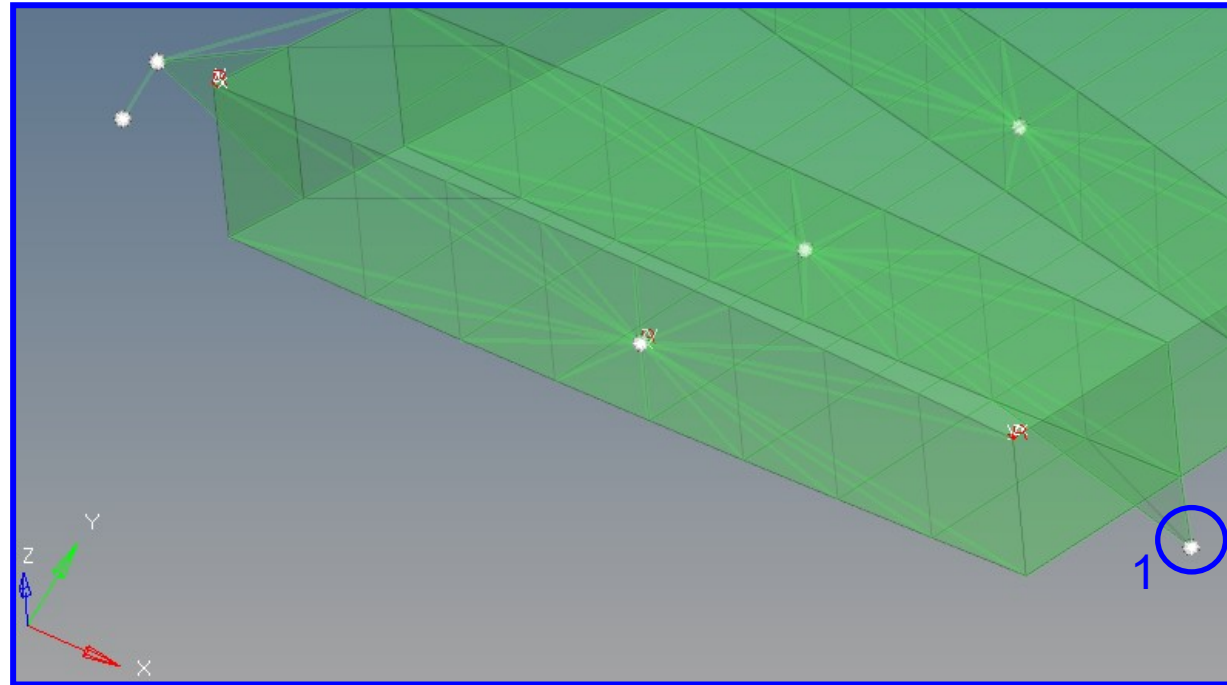
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11 June 2025

REAR FITTING SIZING

From FEM of L10: (Exercise 5)

BOUNDARY CONDITIONS:

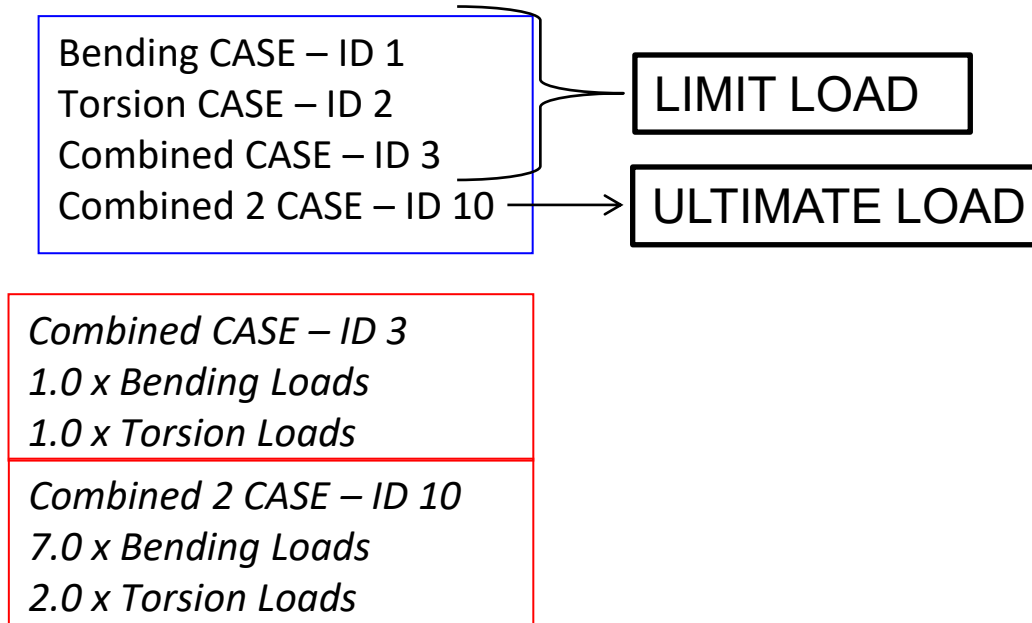


Support conditions – SPCs ID 510:

DOFs 123 constraints in the Pivot Point (node 5402001)

REAR FITTING SIZING

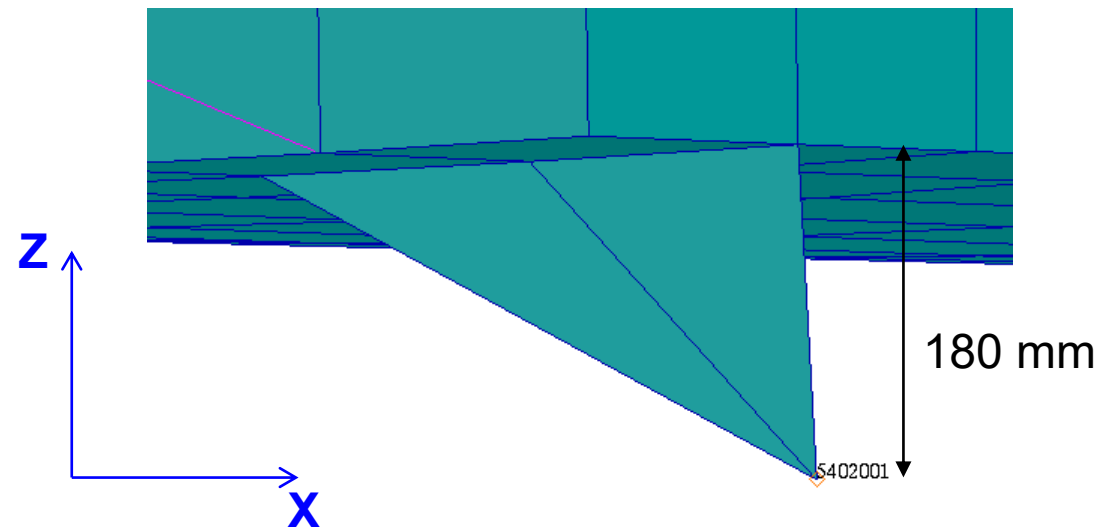
From FEM of L10:

LOAD CASES:

REAR FITTING SIZING

From FEM:

Geometry:



Lug is in XZ-Plane. Assume Rear Fitting is the Female lug

Obtain the SPC loads at rear fitting:

From *.f06:

FORCES OF SINGLE - POINT CONSTRAINT								
POINT ID.	TYPE	T1	T2	T3	R1	R2	R3	
5402001	G	-8.819985E+01	-1.321991E+03	-1.714820E+04	0.0	0.0	0.0	
5402001	G	-5.361471E-10	-2.299440E+01	6.155019E+03	0.0	0.0	0.0	
5402001	G	-8.819985E+01	-1.344985E+03	-1.099319E+04	0.0	0.0	0.0	
5402001	G	-6.173989E+02	-9.299923E+03	-1.077274E+05	0.0	0.0	0.0	

In your model SPC is defined in Global Coordinate System.

REAR FITTING SIZING

Obtain the lug loads at rear fitting:

Lug load is then:

LC ID	Limit Load			$P_{LL} \text{ (N)}$	$P_{y_LL} \text{ (N)}$
	Fx (N)	Fy (N)	Fz (N)		
1	-88	-1322	-17148	17148	-1322
2	0	-23	6155	6155	-23
3	-88	-1345	-10993	10994	-1345
10	-412	-6200	-71818	71819	-6200

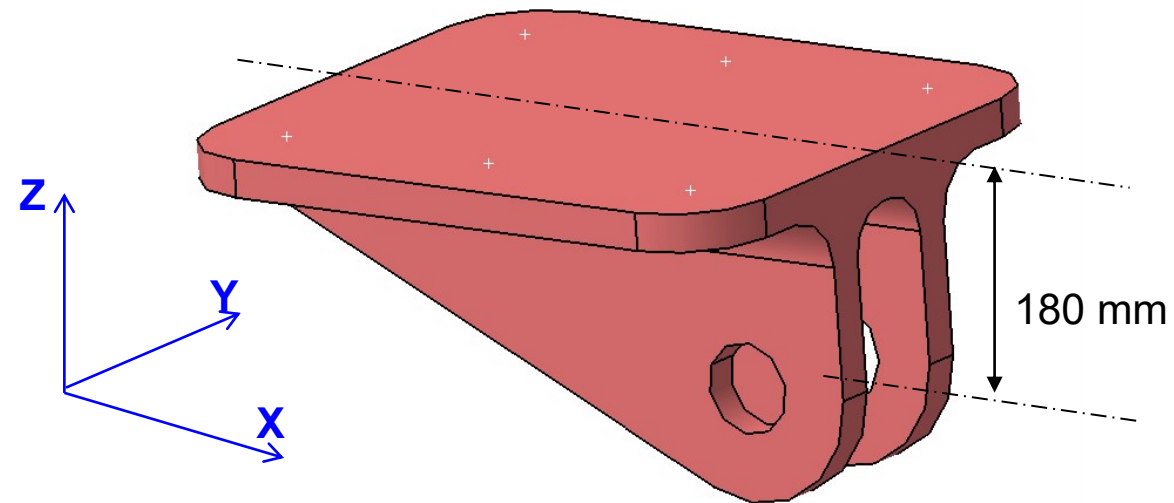
$$P = \sqrt{P_x^2 + P_z^2}$$

With SF = 1.5:

LC ID	Ultimate Load			$P_{UL} \text{ (N)}$	$P_{y_UL} \text{ (N)}$
	Fx (N)	Fy (N)	Fz (N)		
1	-132	-1983	-25722	25723	-1983
2	0	-34	9233	9233	-34
3	-132	-2017	-16490	16490	-2017
10	-617	-9300	-107727	107729	-9300

REAR FITTING SIZING

Fitting sketch:

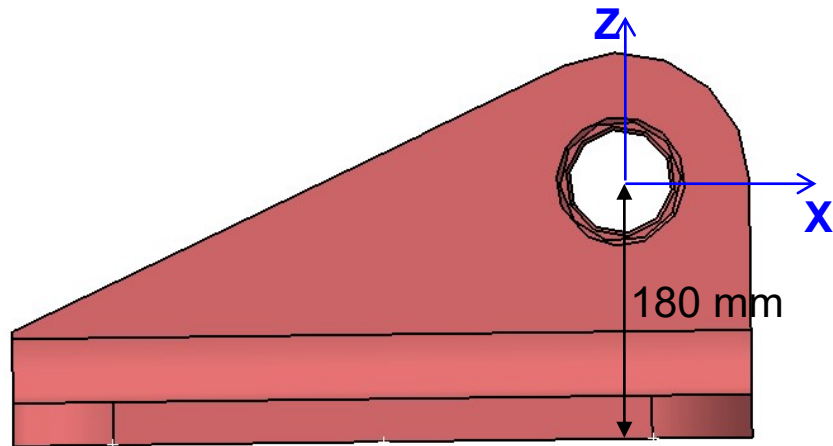


Lug is in XZ-Plane. Assume Rear Fitting is the Female lug

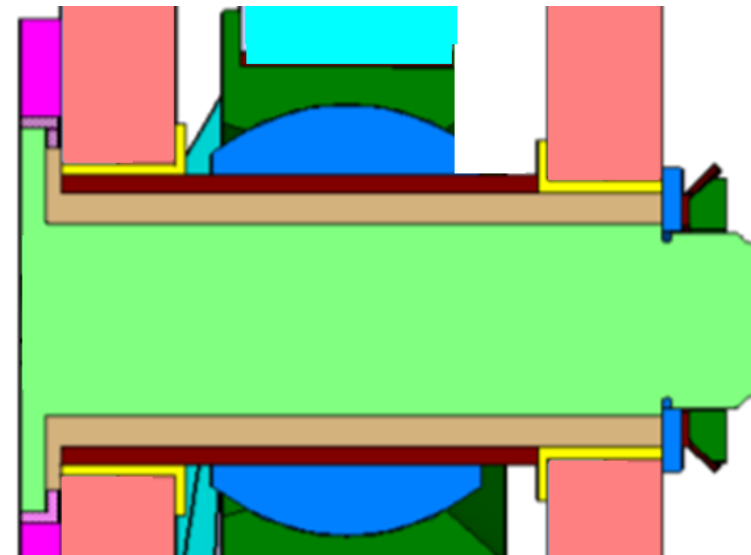
REAR FITTING SIZING

Obtain the lug loads at rear fitting:

Sizing load case: LC 10. With a coordinate system located in your fitting lug:



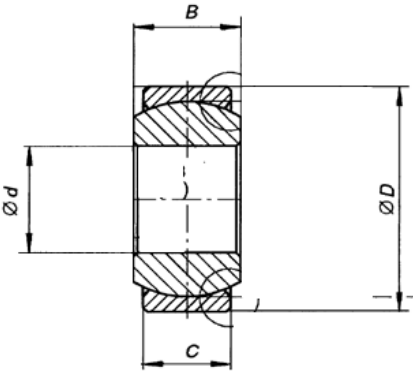
	LC: 10		
	Fx (N)	Fy (N)	Fz (N)
LL	-412	-6200	-71818
UL	-617	-9300	-107727



REAR FITTING SIZING

BEARING ANALYSIS:

From Standard:



Diameter code	d (mm)	D (mm)	B (mm)	C (mm)	Pradial (N)	Paxial (N)
10	15.9	30.2	15.9	12.7	138000	55000

Bearing Analysis		
$P_{Rad, UL} (N)$		107729
BR-10	d (mm)	15.9
	D (mm)	30.2
	C (mm)	12.7
	$P_{Rad} (N)$	138000
	$P_{Ax} (N)$	55000
RF_{Br}	1.28	$P_{Rad, all} / (P_{Rad, UL} \cdot FF)$
	FF	1.0

REAR FITTING SIZING

BUSHING ANALYSIS:

<u>Plain Bushing Analysis</u>			
Material	Bronze		
Compression yield allowable	Fcy (Mpa)	590	
Bush thickness (mm)	t _{bush} (mm)	1.5	Standard value
Female thickness (mm)	t _o (mm)	12	Assumption, first attempt
Inner diameter (mm)	d (mm)	12.9	
Bearing area (mm ²)	Abr (mm ²)	154.8	D-to
Bush load allowable (N)	Pbry (N)	168964	1.85·Fcy·Abr
Applied load (N)	P _{LL} (N)	35910	Assumed g1=g2=g; P1=P2=Pi/2
Reserve Factor	RF _{bush}	4.7	
	FF	1.0	
<u>Flanged Bushing Analysis</u>			
Material	Bronze		
Compression yield allowable	Fcy (Mpa)	590	
Bush thickness (mm)	t _{bush} (mm)	1	Standard value
Female thickness (mm)	t _o (mm)	12	
Inner diameter (mm)	d (mm)	15.9	
Bearing area (mm ²)	Abr (mm ²)	190.8	D-to
Bush load allowable (N)	Pbry (N)	208258	1.85·Fcy·Abr
Applied load (N)	P _{LL} (N)	35910	Assumed g1=g2=g; P1=P2=Pi/2
Reserve Factor	RF _{bush}	5.79	
	FF	1.0	

REAR FITTING SIZING

LUG ANALYSIS

Female Lug Analysis		
LUG GEOMETRY:		
Lug external ratio:	R_0 (mm)	17.9 $1.5 < 2R_0/D < 3$
Lug hole diameter:	D (mm)	17.9 Bolt diameter + 2·(bushes thickness)
Lug width:	W (mm)	35.8 $2 \cdot R_0 / \cos(\beta/2)$
Distance from center hole to external ratio:	a (mm)	17.9 R_0
Thickness female:	t_0 (mm)	12.0
Tapered angle	β (deg)	0 conservatively
Areas	A1 (mm ²)	138.9
	A2 (mm ²)	107.4
	A3 (mm ²)	107.4
	A4 (mm ²)	138.9
	Abr (mm ²)	214.8
	At (mm ²)	214.8
	Aav (mm ²)	126.5
MATERIAL PROPERTIES:		
Lug material:	A17475T7351	
Billet	100	
Ultimate tension load:	Ftu_L [N/mm ²]:	455.1
	Ftu_T [N/mm ²]:	461.9
Yield tension load:	Fty_L [N/mm ²]:	372.3
	Fty_T [N/mm ²]:	372.3
Yield compression load:	Fcy [N/mm ²]:	351.6
Ultimate shear load:	Fsu [N/mm ²]:	282.7
Plastic bending:	F0/Ftu (7075-T6, Bruhn)	70/75
	k	1.5
	Fbu [N/mm ²]:	667.5 $Ftu \cdot (1 + F0/Ftu(7075-T6) \cdot (k-1))$

REAR FITTING SIZING

LUG ANALYSIS (cont)

ALLOWABLE LUG LOADS:		
Axial loads, tensile failure (1a)	Ptu (N)	88469
	W/D	2.0
	Kt	0.91 Graphic, curve 4
	Ftu [N/mm ²]:	455.1
	At (mm ²)	214.8
Axial loads, shear-bearing failure (1b)	Pbru (N)	80502
	a/D	1.0
	D/t	1.5
	Kbr	0.82 Graphic
	Ftu [N/mm ²]:	455.1
Axial loads, yield failure (1c)	Abr (mm ²)	214.8
	Py (N)	72441
	Pu)min/AbrFtu	0.8
	C	1.1 Graphic
	Fty/Ftu	0.8
ALLOWABLE LUG LOADS:		
Transverse loads (2a)	Pu)min (N)	80502
	Ptru (N)	31174
	Aav/Abr	0.6
	Ktru	0.3 Graphic
	Ftu [N/mm ²]:	461.9
Transverse loads, yield failure (2b)	Abr (mm ²)	214.8
	Ptry (N)	55979 Ktry-Fty-Abr
	ktry	0.7
Allowable oblique loads	Fty [N/mm ²]:	372.3
	θ (deg)	0
	P _{uθ} (N)	80444 $1/(\cos\theta/\min(Ptu,Pbru)^{1.6}+(\sin\theta/Ptru)^{1.6})^{0.625}$
	P _{yθ} (N)	72424 $1/(\cos\theta/Py)^{1.6}+(\sin\theta/Pty)^{1.6})^{0.625}$

REAR FITTING SIZING

LUG ANALYSIS (cont)

RESERVE FACTORS:

Ultimate check (3a)

$RF_{lug, UL}$	$1.49 P_{y\delta}/(P_{lug, UL} \cdot FF)$
$P_{y\delta}$ (N)	80444
$P_{lug, UL}$ (N)	53865
FF	1.0

Limit check (3b)

$RF_{lug, LL}$	$2.02 P_{y\delta}/(P_{lug, LL})$
$P_{y\delta}$ (N)	72424
$P_{lug, LL}$ (N)	35910

REAR FITTING SIZING

LUG ANALYSIS (cont)

<u>Cross-section female lug analysis</u>		
GEOMETRICAL DATA		
Distance from center hole to critical section	h (mm)	170 Assumed 10 mm of base thickness
Thickness	t0 (mm)	12.0
Width	width (mm)	206 W+h (engineering assumption)
Area	A (mm ²)	2470 w·t0
Torsion constant	J (mm ⁴)	118541 +1/3·t ³ ·width
LIMIT CHECK		
Limit Loads	Px (N)	-206 Each lug support half
	Py (N)	7182 10% conservatively assumed. Lateral load is supported by only one female lug
	Pz (N)	35909 Each lug support half
Tension stress	σt (Mpa)	14.5 Pz/A
Bending moment	Mx (Nmm)	1220931 Py·h
	My (Nmm)	-3087262 Px·h-Pz·(width/2-R0)
Bending stress	σb (Mpa)	283.6 6·abs(My)/t·width ² +6·abs(Mx)/width·t ²
Torsion	Mz (Nmm)	610465 Py·(width/2-R0)
Shear stress	τ (Mpa)	64.7 Pshear/A+abs(Mz)·t/J
Material Data		Al7475T7351
	Ftu (Mpa)	455.1
	Fty (Mpa)	372.3
	Fcy (Mpa)	351.6
	Fsy (Mpa)	231.3 Fsu·Fty/Ftu
	Fsu (Mpa)	282.7
Reserve Factor bending	RF _{bn,LL}	1.17 min(Fty, Fcy)/(σt+σb)
Reserve Factor shear	RF _{sh,LL}	3.57 Fsy/τ

REAR FITTING SIZING

LUG ANALYSIS (cont)

ULTIMATE CHECK		
Ultimate Loads	Px (N)	-309 Each lug support half
	Py (N)	10773 10% conservatively assumed. Lateral load is supported by only one female lug
	Pz (N)	53864 Each lug support half
Tension stress	σ_t (Mpa)	21.81 Pz/A
Bending moment	Mx (Nmm)	$1831396 \text{ Py} \cdot h$
	My (Nmm)	$-4630893.41 \text{ Px} \cdot h - \text{Pz} \cdot (\text{width}/2 - R_0)$
Bending stress	σ_b (Mpa)	$425.5 \cdot 6 \cdot \text{abs(My)}/t \cdot \text{width}^2 + 6 \cdot \text{abs(Mx)}/\text{width} \cdot t^2$
Torsion	Mz (Nmm)	$915698 \text{ Py} \cdot (\text{width}/2 - R_0)$
Shear stress	τ (Mpa)	$97.1 \text{ Pshear}/A + \text{abs(Mz)} \cdot t/J$
Material Data		Al7475T7351
	Ftu (Mpa)	455.1
	Fsu (Mpa)	282.7
	Fbu (Mpa)	667.5
Tension ratio	Rtn	0.0479
Bending ration	Rbn	0.6374
Shear ratio	Rsh	0.3433
Reserve factor	RF _{UL}	2.88
	FF	1.0

REAR FITTING SIZING

PIN BOLT ANALYSIS:

PIN BOLT ANALYSIS		
BOLT GEOMETRY:		
Bolt diameter:	d (mm)	12.9
Gap:	g (mm)	1 Assumed
Male lug	ti (mm)	12.7 Assumed male lug of 12.7 mm is OK
Female lug	to (mm)	12.0
Pin area	A (mm ²)	130.7
Pin Inertia	I (mm ⁴)	1359.3
Arm	b (mm)	10.2 to/2+gap+ti/4
MATERIAL		
Bolt:	PH13-8 Mo h1050	
Ultimate tension load:	Ftu bolt[N/mm ²]:	1206.7
Yield tension load:	Fty bolt[N/mm ²]:	1137.6
Ultimate shear load:	Fsu bolt [N/mm ²]:	696.6
Yield shear load:	Fsy bolt [N/mm ²]:	656.7 Fsu·Fty/Ftu
Plastic bending:	F0/Ftu (17-7PH, Bruhn)	161/180
	k	1.7
	Fbu [N/mm ²]:	1962
ULTIMATE CHECK		
BENDING ANALYSIS		
Applied ultimate load	PuL (N)	107729
Bending moment	M (Nmm)	548072 P·b/2
Bending stress	σ (Mpa)	2601 M·D/2/I
Reserve Factor bending	RFbn	0.75 Fbu/σ
SHEAR ANALYSIS		
Shear stress	τ (Mpa)	412 P/A/2
Reserve Factor Shear	RFsh	1.69 Fsu/τ
BENDING SHEAR COMBINATION		
Bending - Shear combination	RFpin	0.68 1/FF/sqr((1/RFbn)^2+(1/RFsh)^2)
	FF	1.0

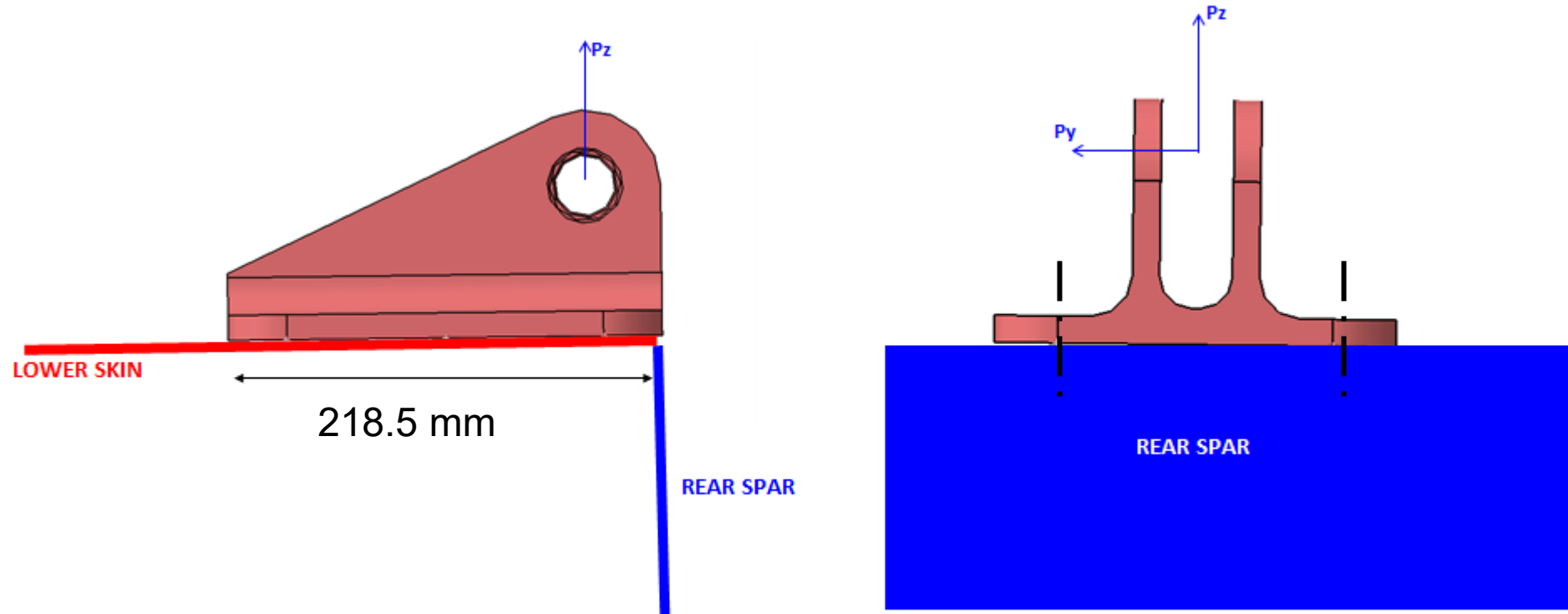
REAR FITTING SIZING

PIN BOLT ANALYSIS (cont):

LIMIT CHECK		
BENDING ANALYSIS		
Applied limit load	PLL (N)	71819
Bending moment	M (Nmm)	365381 P·b/2
Bending stress	σ (Mpa)	1734 M·D/2/I
	Fty bolt[N/mm^2]:	1137.6
Reserve Factor bending	RFbn	0.65 Fty/ σ
SHEAR ANALYSIS		
Shear stress	τ (Mpa)	275 P/A/2
	Fsy bolt [N/mm^2]:	656.7
Reserve Factor Shear	RFsh	2.39 Fsy/ τ

REAR FITTING SIZING

FITTING SCKETCH (Exercise 6: Lug attachment)



Fastener bolt diameter = 6,35 mm

Number of bolts = $(218.5 - 4,5d) / (\text{pitch} = 4,5d) = 6$ bolts/row

Pitch = $(218,5 - 4,5 d) / 6 = 32,2 = 5D$

FITTING SCKETCH

OFFPLANE BOLTGROUP

Due to Pz:

$$P_z = P_{z_app} / 6 \text{ bolts} / 2 \text{ rows} = 107727 / 12 = 8977 \text{ N}$$

Due to My: $M_y = P_z \cdot (\text{width}/2 - W/2) = 107727 \cdot (218,5/2 - 35,8/2) = 9840861 \text{ Nmm}$

$$P_z = M_y / (\text{width} - 4d) = 50962 \text{ N}$$

Due to Mx: $M_x = P_y \cdot 180 = 9300 \cdot 180 = 1674000 \text{ N}$

$$P_z = M_x / \text{distance_between_rows} / 6 = 1674000 / 69,4 / 6 = 4020 \text{ N}$$

$$\text{distance between rows} = 69,4 \text{ mm} (12,7 + 4 + 12 \cdot 2 + 2 \cdot 6,35 + 2 \cdot 8 \text{ (radio)})$$

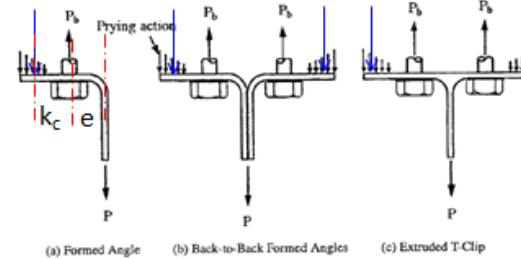
Total tension load: $P_z = 8977 + 50962 + 4020 = 63959 \text{ N}$

REAR FITTING SIZING

FITTING SKETCH

Prying:

- Prying factor shall be taken into account when eccentric loads are applied on the fastener.



$$P_{bolt} = P \cdot C_{prying}; \text{ where } C_{prying} = 1 + \frac{e}{k_c}$$

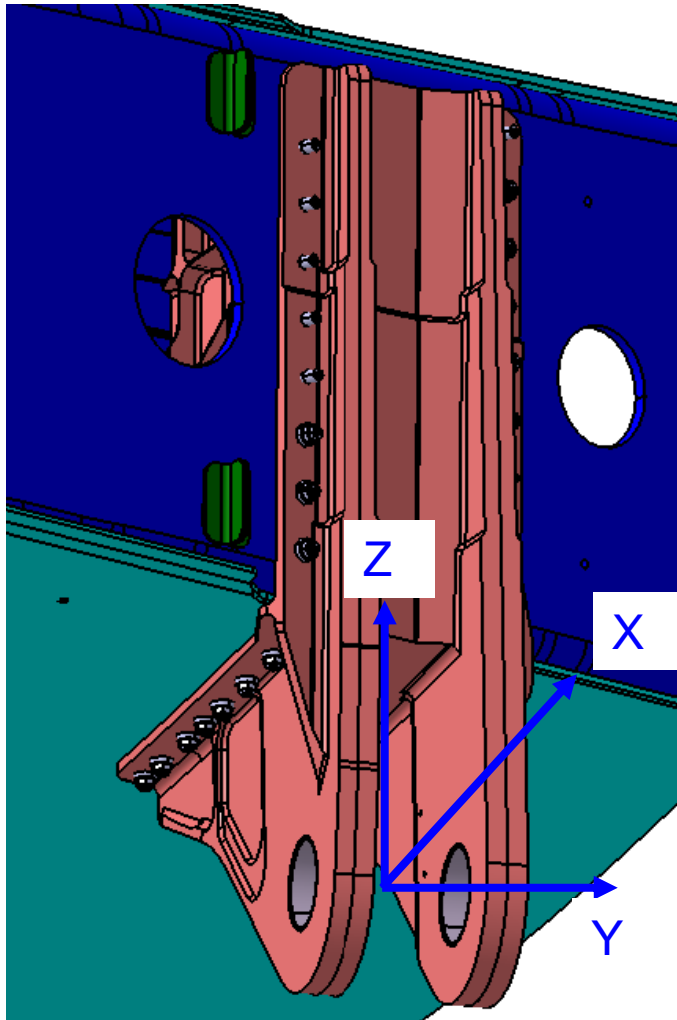
e (mm) = half of the female thickness + D + Radio = $12/2 + 6,35 + 8 = 20.4$ mm

K_c (mm) = $2/3 \cdot \text{edge distance} = 2/3 \cdot 2 \cdot 6,35 = 8.5$ mm

$$\text{Due to prying: } P_z = 63959 \cdot \left(1 + \frac{20.4}{8.5}\right) = 63959 \cdot (1 + 2.4) = 217461N$$

REAR FITTING SIZING

FITTING SCKETCH



INPLANE BOLTGROUP

P_z : 107727 N

With only 2 rows
And 6 fasteners per row

$$P_z = 107727/6/2 = 8977 \text{ N}$$

P_y : Assuming all load is transferred to the spar

$$P_y = 9300/6/2 = 775 \text{ N}$$

Assuming M_x that it is generated by the P_y is supported by the contact with the skin

$$P_{sh_spar} = 9752 \text{ N with no tension}$$

$$P_{tn_skin} = 4020 \cdot (1+2,4) = 13668 \text{ N}$$